

Linda Mhalla

Scientist

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DOB: January 22, 1990

Professional experience

- Jun 2022–now **Scientist**, *Institute of Mathematics, EPFL, Switzerland.*
- Sep 2020–Jun 2022 **Senior researcher**, *Department of Operations, HEC Lausanne, Switzerland.*
- Sep 2018–Aug 2020 **Postdoctoral researcher**, *Department of Decision Sciences, HEC Montreal, Canada.*
- 2014–2018 **Teaching assistant**, *Geneva School of Economics and Management, University of Geneva, Switzerland.*
- Sep 2013–Jan 2014 **Intern in statistical analysis of ozone data**, *Federal Office of Meteorology and Climatology MeteoSwiss, Switzerland.*

Education

- 2014–2018 **PhD in Statistics**, University of Geneva, Switzerland.
Statistical modelling and inference for covariate-dependent extremal dependence.
Supervised by Prof. Valérie Chavez-Demoulin and Prof. Elvezio Ronchetti.
- 2012–2014 **MSc in Mathematical Engineering**, Ecole Polytechnique Fédérale de Lausanne (EPFL), Switzerland.
Master thesis: Joint estimation for extremes of aggregated data.
Supervised by Prof. Anthony C. Davison.
- 2009–2012 **BSc in Mathematics**, Ecole Polytechnique Fédérale de Lausanne (EPFL), Switzerland.
- 2008–2009 **Special Mathematics course (CMS, preparatory class)**, Ecole Polytechnique Fédérale de Lausanne (EPFL), Switzerland.

Research grants

- Sep 2020–May 2021 **Swiss National Science Foundation, Postdoc.Mobility fellowship**
- Mar 2019–Sep 2020 **Swiss National Science Foundation, Early Postdoc.Mobility fellowship**
- Sep 2018–Mar 2019 **Statlab-INCASS-CRM postdoctoral fellowship**
- 2017 **Travel grant from Swiss Academy of Humanities and Social Sciences**
- 2016 **Travel grant from Société Académique de Genève**
- 2016 **Travel grant from Fondation Ernst et Lucie Schmidheiny**

Academic service

- 2022 **Co-organizer of the the Banff workshop on "Combining Causal Inference and Extreme Value Theory in the Study of Climate Extremes and their Causes"**
- 2015–2018 **Representative of the PhD candidates** at Conférence Universitaire de Suisse Occidentale (CUSO)

2017 Co-organizer of the CUSO Young Researchers Conference

Teaching experience and supervision of junior researchers

Feb 2023–Now Mathematics Section, EPFL, Switzerland

Lecturer:

- "Probabilités et statistique" for Bachelor students
- "Risk and environmental sustainability" for Bachelor students
- "Statistical computation and visualisation" for Master students
- "Applied statistics" for Master students
- "Bioinformatic analysis of RNA-sequencing" for Doctoral students

Student supervision:

- Semester project in statistics (2)
- Master thesis (8)

Feb 2021–Jun 2021 Department of Operations, HEC Lausanne, Switzerland

Lecturer: "Statistiques II" for Bachelor students

Mar 2020–Dec 2020 Co-supervisor with Prof. Marco Bee (University of Trento) and Prof. Valérie Chavez-Demoulin (HEC Lausanne) of Fabio Fusato, MSc student in Mathematics

Mar 2020–Sep 2020 Co-supervisor with Prof. Valérie Chavez-Demoulin and Eric Jondeau (HEC Lausanne) of Kevin Billy, MSc student in Finance

Dec 2019–Dec 2021 Co-supervisor with Prof. David Ardia (HEC Montréal) of Jaime Casigay, MSc student in Financial Engineering

Jan 2020–Apr 2020 Department of Decision Sciences, HEC Montréal, Canada

Lecturer: "Analyse et inférence statistique" for Master students in Management

2014–2018 Geneva School of Economics and Management, University of Geneva, Switzerland

Teaching assistant:

- "Multivariate analysis" for Master students in Statistics and Economics
- "Sampling techniques" for Master students in Statistics and Economics
- "Probability and statistics" for Bachelor students in Economics and Management
- "Mathematics II" for Bachelor students in Economics and Management

2009–2014 Ecole Polytechnique Fédérale de Lausanne (EPFL), Switzerland

Teaching assistant:

- "Mathematical analysis" for Bachelor students in Mechanical Engineering and Electrical Engineering
- "Mathematical analysis" for students in Polymaths (Preparatory Course)
- "Mathematical analysis" for students in CMS (Preparatory Course)

Publications

- 2025 L. Gimeno-Sotelo, J. Richards, A. Hazra, **L. Mhalla**, and P. de Zea Bermudez. A review of applications of extreme value theory to environmental risk assessment, in A. Bekker, J. Ferreira, and P. Nagar (Eds.) *Environmental Statistics: Innovative Methods and Applications*. Taylor & Francis

- 2025 **L. Mhalla**, V. Chavez-Demoulin, and P. Naveau. Causal discovery in multivariate extremes: A study of Swiss hydrological catchments. *Environmetrics* **36**, 36:e70034
- 2024 V. Chavez-Demoulin and **L. Mhalla**. Causal inference for extreme values, in M. de Carvalho, R. Huser, P. Naveau, and B.J. Reich (Eds.) *Handbook on Statistics of Extremes*. Chapman & Hall / CRC
- 2021+ V. Chavez-Demoulin, E. Jondeau, and **L. Mhalla**. Climate-related disasters and death toll. Available at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3918201
- 2021 **L. Mhalla**, J. Hambuckers, and M. Lambert. Extremal connectedness of hedge funds. *Journal of Applied Econometrics* **37**, 988–1009
- 2021 D. Castro-Camilo, **L. Mhalla** and T. Opitz. Bayesian space-time gap filling for inference on extreme hot-spots: an application to Red Sea surface temperatures. *Extremes* **24**, 105–128
- 2020 **L. Mhalla**. Discussion of “Graphical models for extremes” by Sebastian Engelke and Adrien S. Hitz. *Journal of the Royal Statistical Society: Series B* **82**, 871–932.
- 2020 **L. Mhalla**, V. Chavez-Demoulin, and D.J. Dupuis. Causal mechanism of extreme river discharges in the upper Danube basin network. *Journal of the Royal Statistical Society: Series C* **69**, 741–764.
- 2019 **L. Mhalla**, M. de Carvalho, and V. Chavez-Demoulin. Regression type models for extremal dependence. *Scandinavian Journal of Statistics* **46**, 1141–1167.
- 2019 **L. Mhalla**, T. Opitz, and V. Chavez-Demoulin. Exceedance-based nonlinear regression of tail dependence. *Extremes* **22**, 523–552.
- 2017 **L. Mhalla**, V. Chavez-Demoulin, and P. Naveau. Non-linear models for extremal dependence. *Journal of Multivariate Analysis* **159**, 49–66.

Refereeing for peer-reviewed journals

Journal of the American Statistical Association
 Journal of the Royal Statistical Society: Series A
 Journal of the Royal Statistical Society: Series C
 Environmetrics
 Annals of Applied Statistics
 Computational Statistics and Data Analysis
 ESAIM: Probability and Statistics
 Scandinavian Journal of Statistics
 Statistics and Computing

Oral presentations at conferences and seminars

- May 2026 **Invited talk**, Colloque International Statistique et Econométrie (Mahdia, Tunisie).
- March 2026 **Invited virtual talk**, The Glasgow-Edinburgh Extremes Network: Thematic session on Causality for extremes (Scotland, UK).
- November 2025 **Invited talk**, Workshop Statistics and Data Science for Climate and Weather Extremes, Karlsruhe Institute of Technology (Karlsruhe, Germany).
- October 2025 **Invited virtual talk**, Workshop SGT methods, Institut des actuaires (France).

- June 2025 **Invited talk**, 14th International Conference on Extreme Value Analysis (North Carolina, USA).
- Feb 2025 **Invited talk**, Workshop Extreme events in economics, finance, and society (Trento, Italy).
- Feb 2024 **Invited talk**, Workshop Causality in Extremes (Geneva, Switzerland).
- Sep 2022 **Invited talk**, Centre International de Rencontres Mathématiques (Marseille, France).
- Feb 2022 **Invited virtual talk**, CLIMACT seminar series.
- Jun 2021 **Invited virtual talk**, 11th International Conference on Extreme Value Analysis.
- Oct 2020 **Invited virtual talk**, "Meet a GERAD researcher!" seminars.
- Mar 2020 **Invited talk**, Department of Decision Sciences, HEC Montréal (Montréal, Canada).
- Dec 2019 **Invited talk**, 12th International Conference on Computational and Methodological Statistics (ERCIM) (London, United Kingdom).
- Oct 2019 **Invited talk**, Fall Meeting of Statlab, University of Montréal (Montréal, Canada).
- July 2019 **Contributed talk**, 11th International Conference on Extreme Value Analysis (Zagreb, Croatia).
- June 2019 **Invited talk**, Operations Research Seminar, HEC Lausanne (Lausanne, Switzerland).
- June 2019 **Invited talk**, Workshop on Causality and Extremes, EPFL (Lausanne, Switzerland).
- Apr 2019 **Invited talk**, Département de mathématiques, Université de Sherbrooke (Québec, Canada).
- Feb 2019 **Invited talk**, Department of Mathematics and Statistics, Université Laval (Québec, Canada).
- Dec 2018 **Invited talk**, 11th International Conference on Computational and Methodological Statistics (ERCIM) (Pisa, Italy).
- Dec 2017 **Invited talk**, 10th International Conference on Computational and Methodological Statistics (ERCIM) (London, United Kingdom).
- Jun 2017 **Contributed talk**, 10th International Conference on Extreme Value Analysis (Delft, the Netherlands).
- Jun 2017 **Internal seminar**, Young Researchers Conference (Lausanne, Switzerland).
- Feb 2017 **Contributed talk**, Workshop on Risk Quantification and Extreme Values in Applications (Lausanne, Switzerland).
- Jun 2016 **Internal seminar**, Young Researchers Conference (Lausanne, Switzerland).
- Dec 2016 **Invited talk**, 9th International Conference on Computational and Methodological Statistics (ERCIM) (Sevilla, Spain).
- Apr 2016 **Invited talk**, Workshop on Uncertainty and Causality Assessment in Modeling Extreme and Rare Events (Boulder, USA).

Languages

Fluent English
 Bilingual French
 Native Arabic